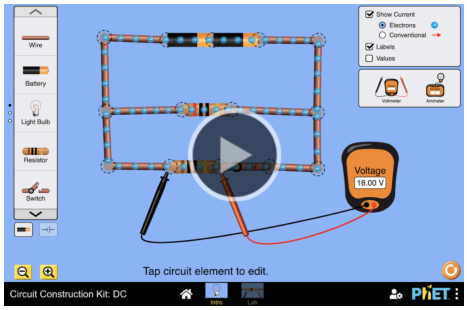
Circuits Lab

Name: _____

To access the PhET simulation type
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Look for the picture to the right.



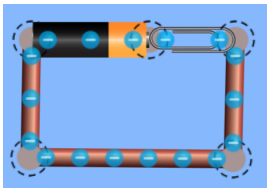
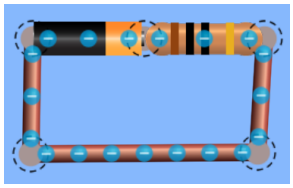
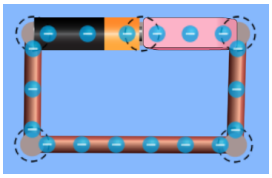
Click the "intro" screen.



I. Engage

1. *A circuit is a closed path that electrons can pass through.* Make and draw 2 circuits.

2. **Remake** the circuits below and compare. Measure the current 

Circuit with a battery & paper clip	Circuit with a battery and resistor	Circuit with a battery and eraser
		
Current:	**Current:**	**Current:**

3. What difference in electrons do you observe between the circuits?

_____

_____

4. Based on your observations, what is "current"?

_____

_____

5. Based on your observations, what does a resistor do?

_____

_____

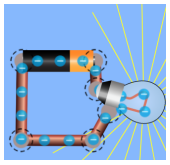
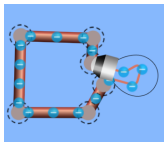
6. Rubber is known as an *insulator*. Based on your observations, what is the definition of an *insulator*?

_____

Circuits Lab

Name: _____

7. **Remake** the *circuits* below. **Measure** the current. **Write** observations about electrons.

Circuit with a battery	Current	Observations about electrons
		_____ _____
		_____ _____

8. Based on your observations, what does the battery do?

_____

_____

Checkpoint. *Check in with your teacher if possible.*

II. Explore

8. Create 3 separate circuits side by side. **Draw** light rays to represent "brightness."

1 battery**, 1 light bulb**, wires	1 battery, **2 light bulbs**, wires	1 battery, **3 light bulbs**, wires
Current:	Current:	Current:

9. What happened to the current as the number of light bulbs increased? _____

10. What else happened as the number of bulbs increased? _____

11. What relationship do you see, based on the last two questions? _____

_____

12. What does a resistor and a light bulb have in common?

_____

Checkpoint. *Check in with your teacher if possible.*

13. *Create 3 new circuits side by side*. Draw bulbs' light rays to represent "brightness." Click "values" and complete the table below.

1 battery, 1 light bulb, wires	**2 batteries**, 1 light bulb, wires	**3 batteries**, 1 light bulb, wires
Voltage:	Voltage:	Voltage:
Resistance:	Resistance:	Resistance:
Current:	Current:	Current:

14. Explain the pattern and/or relationship you see in the data..

_____

_____

_____

15. Mathematically, how could you use Current and Resistance to find the voltage?

_____

16. Mathematically, how could you use Voltage and Resistance to find the current?

_____

17. What might an equation look like to find current? _____

18. What might an equation look like to find voltage? _____

III. Explain. Ohm's Law

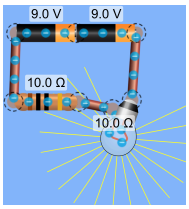
Ohm's Law states that current, I, is dependent on _____ and _____.

Stop here If completing Ohm's Law Lab first.

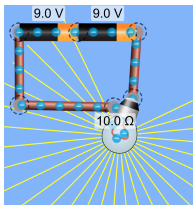
Circuits Lab

Name: _____

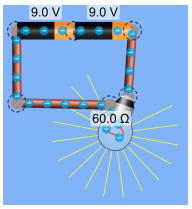
19. Ohm's Law is traditionally written as $V = I \times R$. Use the equation to find current for the circuits:



A. Current: _____



B. Current: _____



C. Current: _____

20. Explain the difference in currents.

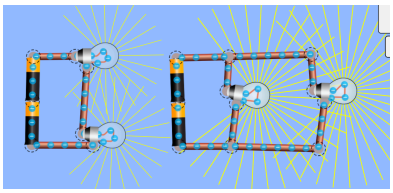
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IV. Extend:

21. **Create circuits 1 & 2 side by side**. Compare the two. What makes them different?

Circuit 1:**series** →



← Circuit 2: **Parallel**

_____

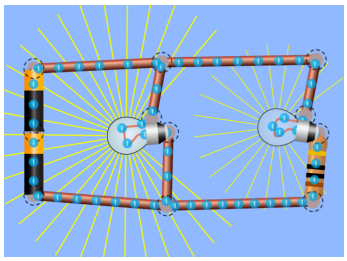
_____

_____

One way to analyze the difference between series and parallel circuits is with the analogy of traffic. Imagine 100 cars on the freeway. The SERIES freeway only has one lane, but the PARALLEL freeway has multiple lanes. Which would you take?
The parallel freeway which allows more cars (electrons) to pass per second. When the lanes diverge, you have one resistor (light bulb) per lane. With a "series" of lightbulbs, all resistors are in one lane.



22. **Explain** the difference in the lightbulbs to the left.



_____

_____

_____

_____

_____

Checkpoint. *Check in with your teacher if possible.*

V. Engineering Design

23. Christmas lights used to be the worst. When one got pulled out or didn't work, the rest would not either. Using your knowledge of types of circuits, explain how Christmas lights were designed and how they might have improved.

_____

_____

_____

Circuits Lab

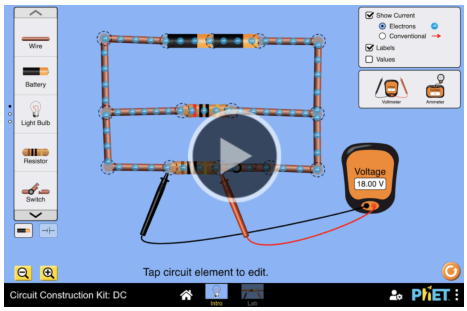
Name: _____

Answer Key

To access the PhET simulation type
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Look for the picture to the right.



Click the "intro" screen.

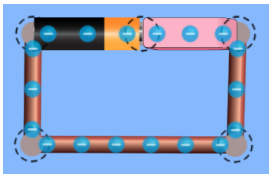
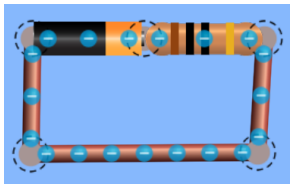
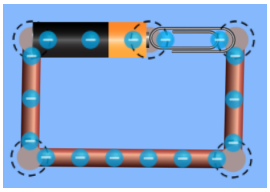


I. Engage

1. *A circuit is a closed path that electrons can pass through.* Make and draw 2 circuits.

Student choice	

2. **Remake** the circuits below and compare. Measure the current 

Circuit with a battery & paper clip	Circuit with a battery and resistor	Circuit with a battery and eraser
		
Current: 0 amps	**Current: .9 amps**	**Current: 89,000 amps**

3. What difference in electrons do you observe between the circuits?

The electrons are moving at different speeds; In the last circuit, the electrons are doing nothing. They are not moving_

4. Based on your observations, what is "current"?

The current is the speed of the electrons

5. Based on your observations, what does a resistor do?

A resistor slows down the electrons and decreases the current.

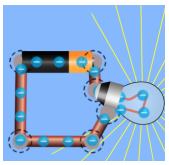
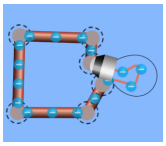
6. Rubber is known as an *insulator*. Based on your observations, what is the definition of an *insulator*?

An insulator is a material that does not allow heat or electrical conduction.

Circuits Lab

Name: _____

7. **Remake** the *circuits* below. **Measure** the current. **Write** observations about electrons.

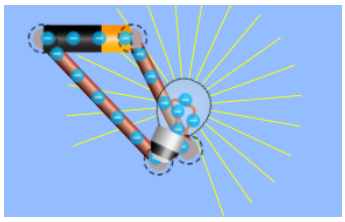
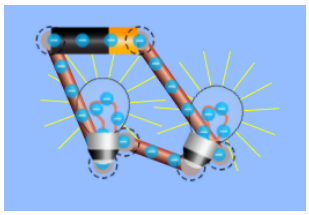
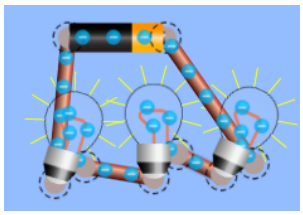
Circuit with a battery	Current	Observations about electrons
	**.9 amps**	**The electrons move slowly**
	**0 amps**	**The electrons don't move**

8. Based on your observations, what does the battery do? **The battery gives the electrons the electromotive force to push the electrons through the circuit.**

Checkpoint. *Check in with your teacher if possible.*

II. Explore

8. Create 3 separate circuits side by side. **Draw** light rays to represent "brightness."

1 battery**, 1 light bulb**, wires	1 battery, **2 light bulbs**, wires	1 battery, **3 light bulbs**, wires
Current:	Current:	Current:
 **bright**	 **Less bright**	 **Dim**

9. What happened to the current as the number of light bulbs increased? **The current decreased**

10. What happened to the bulbs as the number of bulbs increased? **The brightness decreases.**

11. What relationship do you see, based on the last two questions?
The current and brightness are directly related. The current affects the brightness.

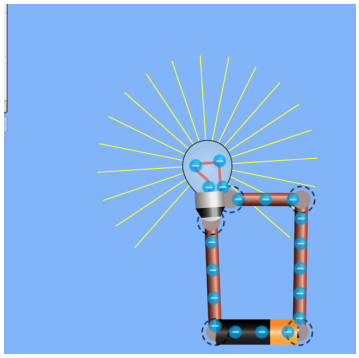
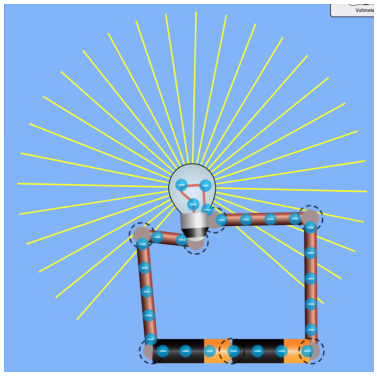
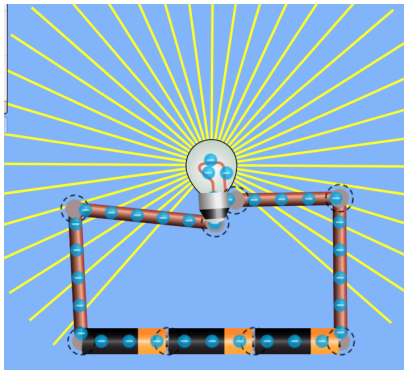
12. What does a resistor and a light bulb have in common?
Light bulbs act as a resistor; they both resist current.

Checkpoint. *Check in with your teacher if possible.*

Circuits Lab

Name: _____

13. *Create 3 new circuits side by side*. Draw bulbs' light rays to represent "brightness." Click "values" and complete the table below.

1 battery, 1 light bulb, wires	**2 batteries**, 1 light bulb, wires	**3 batteries**, 1 light bulb, wires
		
Voltage: **9 volts**	Voltage: **18 volts**	Voltage: **27 volts**
Resistance: **10 ohms**	Resistance: **10 ohms**	Resistance: **10 ohms**
Current: **: .9 amps**	Current: **: 1.8 amps**	Current**: 2.7 amps**

14. Explain the pattern and/or relationship you see in the data..

The brightness and current increases as the voltage does.

15. Mathematically, how could you use Current and Resistance to find the voltage?

Multiply current by resistance.

16. Mathematically, how could you use Voltage and Resistance to find the current?

Divide voltage by resistance.

17. What might an equation look like to find current? ***C = V/R*** ***(technically, I= V/R)***

18. What might an equation look like to find voltage? ***V=C*R*** ***(technically, V=C*R)***

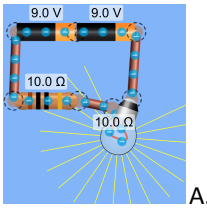
III. Explain. Ohm's Law

Ohm's Law states that current, I, is dependent on **voltage** and **resistance**.

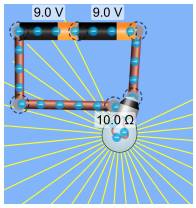
Stop here If completing Ohm's Law Lab first.

Name: _____

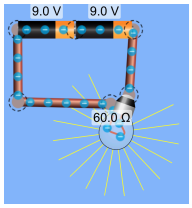
19. Ohm's Law is traditionally written as $V = I \times R$. Use the equation to find current for the circuits:



A. Current: **.9 amps**



B. Current: **1.8 amps**



C. Current: **18/60=.3 amps**

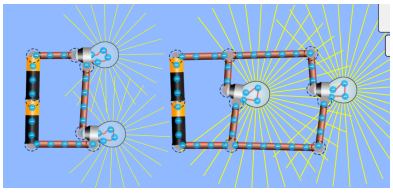
20. Explain the difference in currents.

Current A is lower because the speed of electrons decreases as the resistance increases

IV. Extend:

21. **Create circuits 1 & 2 side by side**. Compare the two. What makes them different?

Circuit 1:**series** →

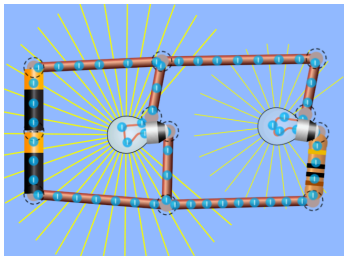


← Circuit 2: **Parallel**

Light Bulbs in parallel are brighter than those in series, which are side by side.

One way to analyze the difference between series and parallel circuits is with the analogy of traffic. Imagine 100 cars on the freeway. The SERIES freeway only has one lane, but the PARALLEL freeway has multiple lanes. Which would you take?
The parallel freeway which allows more cars (electrons) to pass per second. When the lanes diverge, you have one resistor (light bulb) per lane. With a "series" of lightbulbs, all resistors are in one lane.





22. **Explain** the difference in the lightbulbs to the left.

The light bulb with the resistor is less bright than the one without.

Because the resistor is in series and sharing the current of the lightbulb on the right.

Checkpoint. *Check in with your teacher if possible.*

V. Engineering Design

23. Christmas lights used to be the worst. When one got pulled out or didn't work, the rest would not either. Using your knowledge of types of circuits, explain how Christmas lights were designed and how they might have improved.

The lights are in series, which means when one lightbulb is burnt out or comes out, all lights stop working. The Christmas lights should be designed with some lights in parallel so that the others will still work